

INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

GRC101 LAB REPORT

Linux Shell Navigation, File and Directory Management

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Introduction

Governance, Risk and Compliance (GRC) is more than just creating policies and identifying risks. It is also about making sure there is clear evidence to show that the right controls are in place and working effectively. In many organisations, important records such as system logs, configuration files and access records are stored on Linux systems which is why a GRC engineer needs to understand basic Linux commands because it makes it possible to locate, organise and verify the evidence needed during audits, compliance reviews and security investigations.

In this laboratory exercise, I learned how to use the Linux command line to perform basic tasks such as navigating directories, creating and organising files, searching for information, viewing file details and managing file permissions. File permissions enforce access control by ensuring only authorised users can modify sensitive documents, while metadata such as timestamps and ownership provides evidence of when a document was created or changed and by whom, which is valuable during audits to demonstrate integrity and traceability.

Using the **NorthStar Services Ltd.** scenario, I organised governance, risk and compliance documents into a structured evidence folder. Working through the tasks made me realise that keeping files organised is also as important as creating them. If documents are poorly named or stored in the wrong location, it becomes harder to find evidence when it is needed, which can also increase audit delays. Most importantly, I learned that in GRC, saying a control exists is not enough. One should also be able to provide evidence to support it. The Linux skills practised in this lab showed me how proper file management can help achieve that.

Task Evidence

Task 1 – Open the Shell and Identify the Environment

Commands Used

```
whoami  
hostname  
date  
uname -a  
echo $SHELL
```

Screenshot



Figure 1: Linux Environment Information

Result

The commands displayed the current Linux username, VM hostname, system date and time, operating system/kernel information, and the shell currently in use, which confirmed that the virtual machine was ready for the practical.

Recorded Information

- Linux Username: icdfa
- Computer/VM hostname: icdfa
- Current date and time: Fri 17 Jul 2026 05:55:06 WAT
- Operating system/kernel information: Linux icdfa 6.12.13-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.12.13-1kali1 (2025-02-11) x86_64 GNU/Linux
- Shell currently in use: /usr/bin/zsh

Short Explanation

Before carrying out any administrative or security-related task, it is important to identify the Linux environment. Recording the username, hostname, date, OS and shell provides a reference point for the system used throughout the practical.

Task 2 – Create the Main GRC 101 Laboratory Workspace

Commands Used

```
cd ~
mkdir -p GRC101_Lab/{01_Screenshots,02_Command_Log,03_GRC_Evidence_Folder}
ls -l GRC101_Lab

cd ~/GRC101_Lab/03_GRC_Evidence_Folder
mkdir -p Governance Risk Compliance Audit_Evidence Meeting_Records Archive
ls -l
```

Screenshot

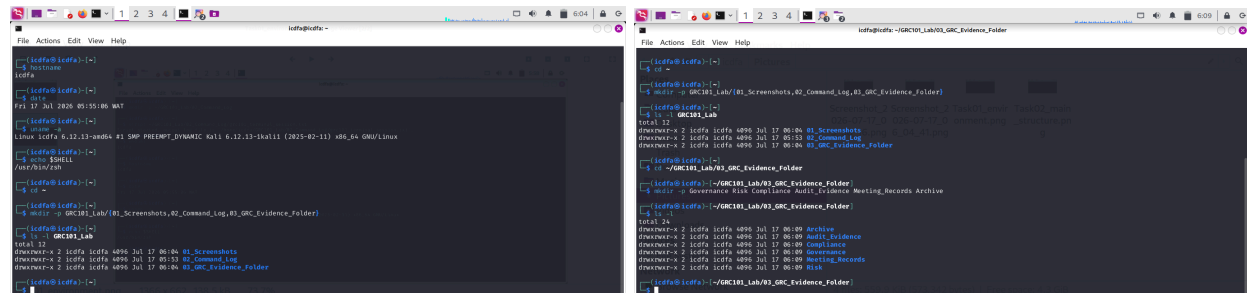


Figure 2: Creation of the Main Laboratory Workspace and GRC Evidence Folder Structure

Result

The required GRC evidence folders were successfully created and organised into separate directories, which makes it easier to manage different categories of records.

Short Explanation

A structured folder hierarchy makes it easier to locate, manage and retrieve evidence. Separating governance, risk, compliance and audit records into different folders keeps documentation effective.

Task 3 – Navigate the Linux File System

Commands Used

```

pwd
cd ~/GRC101_Lab
pwd
cd 03_GRC_Evidence_Folder/Governance
pwd
cd ..
pwd
cd ./Risk
pwd
cd ../Compliance
pwd
cd ~
pwd

```

Screenshot

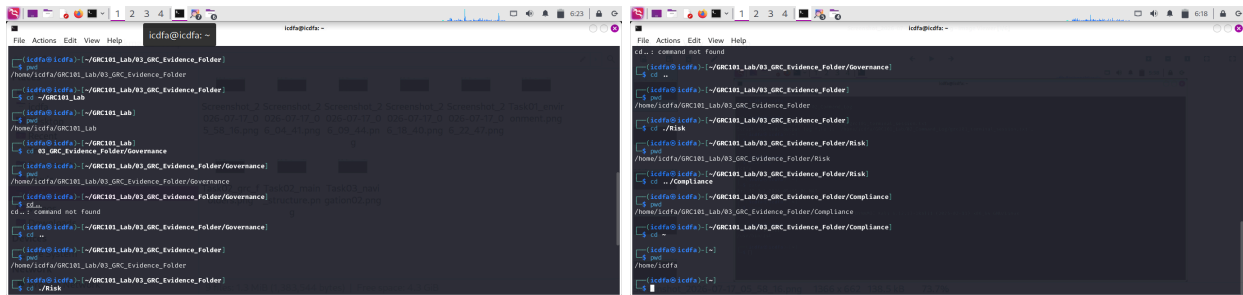


Figure 3: Navigating the Linux File System

Result

The commands successfully navigated between the required directories using both absolute and relative paths.

Short Explanation

- **What pwd does:** It shows the current working directory.
- **The difference between cd .. and cd ~:** cd.. moves to the parent directory, while cd~ returns to the home directory
- **The meaning of . and ...:** The single dot (.) represents the current directory while the double dot (..) represents the parent directory.
- **The difference between an absolute path and a relative path:** Absolute path is the complete address of a file or folder starting from the root of the file system and is represented by /, while a relative path specifies the location of a file or directory based on the current working directory.

Task 4 – List and Inspect Files and Directories

Commands Used

```

cd ~/GRC101_Lab/03_GRC_Evidence_Folder
ls
ls -l
ls -la
ls -lh

echo 'Internal GRC workspace' > .workspace_note
ls -la

```

Screenshot

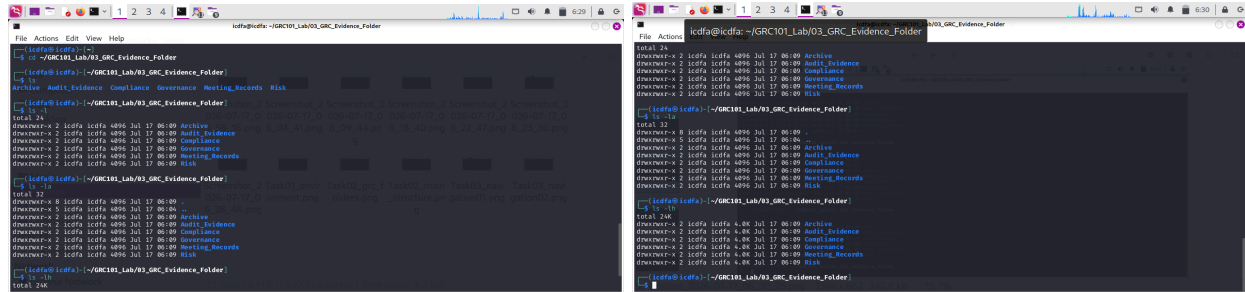


Figure 4: Output of the different ls commands used to display the contents of the GRC evidence folder

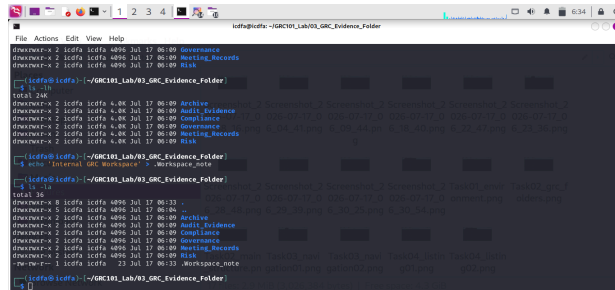


Figure 5: The hidden file .workspace_note after it was created and displayed using the appropriate ls command.

Result

The ls commands displayed the folder contents in different formats. The ls -la command then confirmed the hidden file .workspace_note was created successfully.

Short Explanation

- **Which command displays hidden files?** The command to display hidden files in a terminal is ls -a on Linux
- **What information does ls -l display?** The ls -l command displays a detailed, long-format list of files and directories.
- **Why might a GRC team use hidden configuration files?** A GRC team may use hidden configuration files to store sensitive security settings out of plain sight, support automated compliance scanning, and reduce the risk of accidental modification, which will help protect the confidentiality and integrity of important records.

Task 5 – Create and View GRC Files

Commands Used

```
cd ~/GRC101_Lab/03_GRC_Evidence_Folder
```

```
touch Governance/information_security_policy.txt
```

```
touch Risk/risk_register.txt
```

```
touch Compliance/compliance_checklist.txt
```

```
echo "NorthStar Services Ltd - Information Security Policy" > Governance/information_security_policy.txt
```

```
echo "Owner: GRC Department" >> Governance/information_security_policy.txt
```

```
echo "Review Cycle: Annual" >> Governance/information_security_policy.txt
```

```
echo "Risk ID | Risk Description | Owner | Status" > Risk/risk_register.txt
```

```
echo "R-001 | Unauthorised access to records | IT Manager | Open" >> Risk/risk_register.txt
```

```
echo "Control | Requirement | Status" > Compliance/compliance_checklist.txt
```

```
echo "Access Review | User access reviewed quarterly | Pending" >> Compliance/compliance_checklist.txt
```

```
cat Governance/information_security_policy.txt
cat Risk/risk_register.txt
cat Compliance/compliance_checklist.txt
head -n 2 Governance/information_security_policy.txt
```

Screenshot

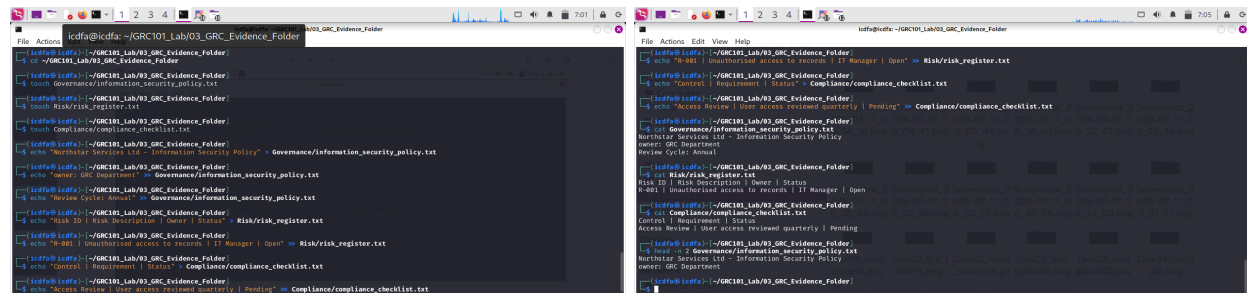


Figure 6: Creation of the required GRC files using the touch and echo commands (left), and verification of the file contents using the cat and head commands (right).

Result

The contents of each file were displayed successfully using cat and head. This confirmed that the information had been saved correctly.

Short Explanation

- **What does the touch command do?** The touch command creates an empty file or update timestamp.
- **What is the difference between > and >>?** The > operator creates a new file or replaces the existing content of a file, while >> adds new content to the end of an existing file without removing what is already there.
- **What is the purpose of cat command?** The cat command displays file content.
- **What is the purpose of head command?** The head command displays the first lines of a file.

Task 6 – Create and View GRC Files

Commands Used

```
cd ~/GRC101_Lab/03_GRC_Evidence_Folder
cp Governance/information_security_policy.txt Audit_Evidence/
cp Risk/risk_register.txt Archive/risk_register_backup.txt
mv Compliance/compliance_checklist.txt Compliance/grc_compliance_checklist.txt
mv Meeting_Records Archive/Old_Meeting_Records
ls -l Audit_Evidence Archive Compliance
```

Screenshot

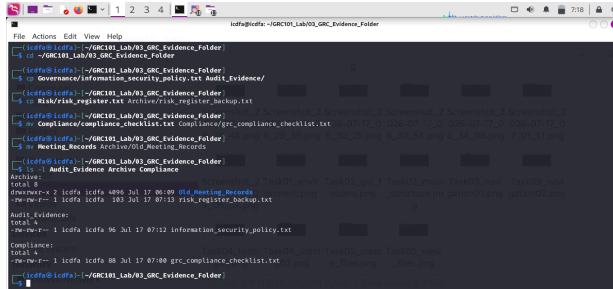


Figure 7 :The successful copying, moving and renaming of files and directories within the GRC evidence repository.

Result

The policy file was copied to Audit_Evidence, the risk register was moved and renamed in Archive, and Meeting_Records was moved into Old_Meeting_Records. The final directory listing confirmed everything was organised correctly.

Short Explanation

- **What is the difference between cp and mv?** The cp copies files or directories, while mv removes a file.
- **State which command you would use to keep an original file while creating a duplicate for an auditor.** I would use the cp command because it preserves the original file and creates an identical copy for review.

Task 7– Search for Files and Content

Commands Used

```
cd ~/GRC101_Lab/03_GRC_Evidence_Folder
find . -type f
find . -type f -name "*.txt"
find . -type f -iname "*policy*"
grep -R "Owner" .
grep -Rni "access" .
```

Screenshot

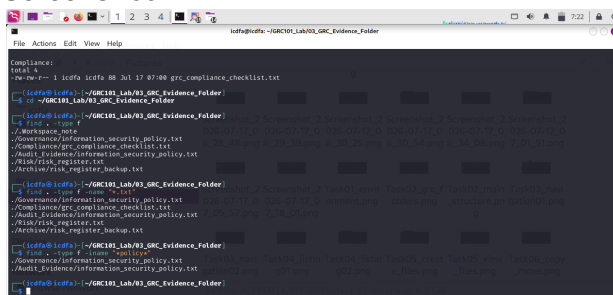


Figure 8: The successful use of the find commands to locate files and search for specific text within GRC documents.

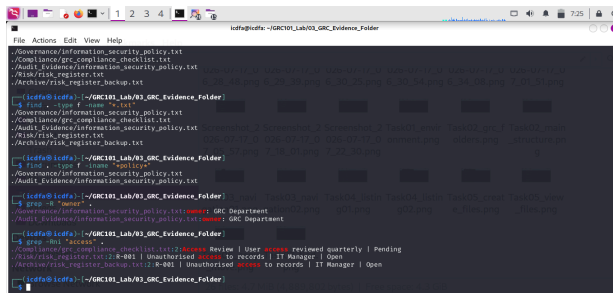


Figure 9: The successful use of the grep commands to locate files and search for specific text within GRC documents.

Result

The find command successfully located the risk_register_backup.txt file, while the grep commands searched the policy and compliance files and displayed the lines containing the specified keywords.

Short Explanation

The find command is used to locate files or directories based on specified search criteria, while the grep command searches within a file for lines containing a particular word or phrase. In this task, find was used to locate a file, whereas grep was used to retrieve specific information from the contents of GRC documents.

Task 8– View File Metadata and Inode numbers

Commands Used

```

cd ~/GRC101_Lab/03_GRC_Evidence_Folder
ls -li Governance/information_security_policy.txt
stat Governance/information_security_policy.txt
df -h
df -i

```

Screenshot

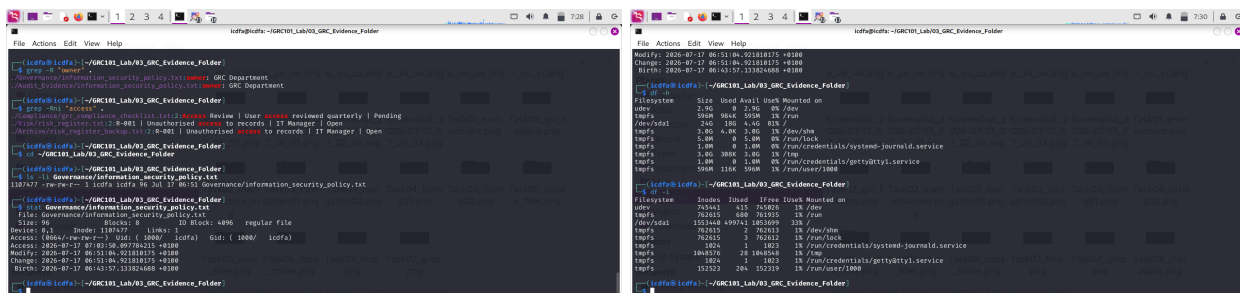


Figure 10: Showing the metadata and inode information for the information_security_policy.txt file.

Result

The commands displayed the metadata of the policy file, including its inode number, owner, file size and last modification time.

File Information

- **Inode Number:** 1107477
- **Owner:** icdfa
- **File Size:** 96 bytes
- **Last Modification Time:** 2026-07-17 06:51:04.921810175 +0100

Short Explanation

File metadata such as ownership, size and modification history helps verify the authenticity of a document. It also makes it traceable.

Task 9– Manage File Permissions

Commands Used

```
cd ~/GRC101_Lab/03_GRC_Evidence_Folder
ls -l Governance/information_security_policy.txt
chmod 640 Governance/information_security_policy.txt
ls -l Governance/information_security_policy.txt
```

Screenshot

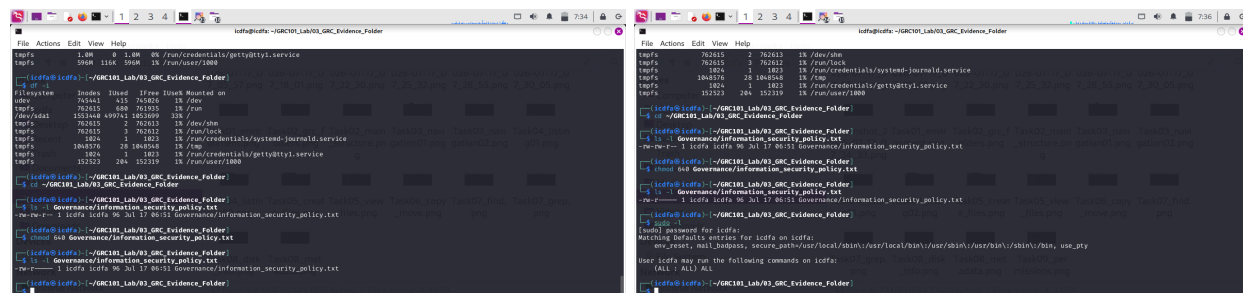


Figure 11: Showing the file permissions before and after modification, as well as the verification of sudo privileges.

Result

The file permissions were successfully changed from their original settings to 640, restricting access based on user roles. The sudo -l command also confirmed the available administrative privileges for the current user.

Short Explanation

Explain what the owner, group and others can do after chmod 640. Do not change system files and do not run sudo commands beyond sudo -l for this task.

After applying chmod 640, the owner has read and write permissions (rw-), the group has read-only permission (r--), and others have no permissions (---). This helps protect sensitive files by ensuring that only authorised users have the appropriate level of access.

Task 10– Delete Files and Directories

Commands Used

```
cd ~/GRC101_Lab/03_GRC_Evidence_Folder
mkdir Temporary
touch Temporary/test_file.txt
ls -la Temporary
rm Temporary/test_file.txt
rmdir Temporary
ls -l
```

Screenshot

Figure 12: Showing the deletion of the temporary file and directory, followed by commands used to confirm that they were successfully removed.

Result

The temporary file and directory were successfully deleted using the appropriate Linux commands. The verification commands then confirmed both items no longer existed.

Short Explanation

The `rmdir` command only works when a directory is empty because it is designed to remove directories that contain no files or subdirectories. This helps prevent the accidental deletion of important data that may still be stored inside a directory.

Task 11– Create a README File Using Nano

Commands Used

```
nano ~/GRC101_Lab/03_GRC_Evidence_Folder/README.txt
cat ~/GRC101_Lab/03_GRC_Evidence_Folder/README.txt
```

Screenshot

Figure 13: Showing the creation of the README.txt file using the Nano text editor.

Result

The README.txt file was successfully created and the required information was entered. The file was saved and Nano was exited successfully.

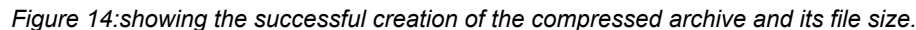
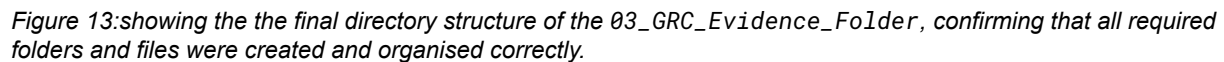
Short Explanation

To save changes in Nano, press **Ctrl + O**, then press **Enter** to confirm the file name. To exit the editor, press **Ctrl + X**. These shortcuts allow changes to be saved before closing the editor.

Task 12– Review the Final Folder Tree and Compress the Work

Commands Used

Screenshot



The tree command confirmed all required GRC directories and files were present. The laboratory workspace was then compressed into the file **GRC101_C11-26-CGRCE-17219.tar.gz**, and its size was verified using the ls -lh command.

The tree command gives a clear view of the directory structure, making it easy to verify everything was created correctly. The tar -czvf command then compresses the whole workspace into one archive for submission.

This exercise showed me that effective GRC is not only about creating policies but also about managing them properly. Creating well-structured folders and naming consistently makes it easier to locate the correct version of a document.

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Challenges and Resolution

During the laboratory exercise, I encountered a few challenges while using Linux commands for the first time. Some commands initially returned errors because of typing mistakes, such as missing spaces, incorrect command names and incorrect folder paths. I also had some difficulty saving files in Nano and transferring screenshots from the Kali VM to my Windows computer.

I resolved these by carefully checking syntax and file paths before re-running commands, and by configuring the shared folder to transfer screenshots. This improved my confidence and reinforced the importance of attention to detail on the command line.

Conclusion

This exercise gave me practical experience navigating the Linux file system, organising documents, managing permissions and creating archives. More importantly, it showed me how these skills support governance and compliance by keeping evidence organised, secure and reliable.

Short Reflection Questions

1. Why is a clear folder structure important during an audit? A clear folder structure makes it easier to locate documents and evidence quickly. It also reduces confusion and helps ensure that important records are organised and easy to verify.

2. How can incorrect file names or locations affect compliance evidence? Incorrect file names or storing files in the wrong location can make important evidence difficult to find. It can create confusion and affect an organisation's ability to demonstrate compliance.

3. What is the difference between copying and moving a file? Copying a file creates a duplicate while keeping the original file unchanged. Moving a file transfers it to another location or renames it, so it no longer remains in its previous location.

4. Why should administrative access through sudo be limited? Administrative access should be limited to authorised users because it allows changes to important system files and settings. Restricting sudo access helps prevent accidental mistakes and reduces security risks.

5. How do file permissions support confidentiality and accountability? File permissions ensure that only authorised users can view or modify sensitive information, which helps protect confidentiality. They also make it easier to determine who is responsible for accessing or changing files, which supports accountability.

6. What evidence from this lab would you present to show that a GRC folder was created correctly? I would present the screenshots showing the folder structure, the output of the tree command, and the compressed archive of the completed laboratory workspace. Together, these demonstrate that the required GRC folders and files were created and organised correctly.